

A Block-Encoded Carleman-Lattice Boltzmann Solver for the Convecting Taylor-Green Vortex

Concept Proposal for the Airbus Global Quantum + AI Challenge 2026

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Abstract

Accurate aerodynamic prediction requires solving nonlinear, dissipative partial differential equations at resolutions whose computational and memory costs grow rapidly with Reynolds number. This paper proposes a fault-tolerant quantum algorithm for the two-dimensional convecting Taylor-Green vortex based on a D2Q9 lattice Boltzmann discretization, Carleman linearization of the nonlinear BGK collision, unitary block encoding of the resulting non-unitary map, and reversible periodic streaming. The benchmark is separated into a periodic box length $L_{\text{box}} = 2\pi$ and vortex scale $L_c = 1$ to remove an ambiguity in the challenge statement. Preliminary implementation validates a local 90-dimensional lifted collision through an eight-qubit unitary dilation with maximum block error 6.94×10^{-17} and maximum unitarity error 5.11×10^{-15} . A full spatial collision-streaming comparison gives velocity relative L_2 error 1.28×10^{-5} between the Carleman and exact BGK paths for a small test. The measured dense routed synthesis is expensive, and the raw post-selection probability is only 1.77%; these results define the optimization target rather than establish advantage. We therefore specify an end-to-end, fixed-accuracy protocol that includes preparation, amplification, routing, fault tolerance, and observable extraction. The target claim is a bounded advantage for selected observables, not full-field recovery.

Keywords: quantum computational fluid dynamics, lattice Boltzmann method, Carleman linearization, block encoding, Taylor-Green vortex, fault-tolerant quantum computing.

1 Motivation and Objective

Airbus identifies predictive aerodynamics as a high-value computational capability because extreme operating conditions remain costly to reproduce in physical tests and expensive to resolve numerically [1]. The challenge asks participants to simulate the convecting Taylor-Green vortex and quantify how runtime, memory or qubits, and numerical error scale with Reynolds number. The exact solution makes this problem unusually suitable for separating physical fidelity from circuit executability.

The proposed method follows the fault-tolerant quan-

tum computing (FTQC) and lattice Boltzmann route. Its research question is:

Can a structured block encoding of collision, combined with logarithmic-size position registers and unitary streaming, reduce time-to-solution or memory for selected flow observables at a fixed physical error as Re increases?

This is intentionally narrower than claiming exponential advantage for an entire velocity field. Recent end-to-end analysis shows that Carleman convergence, time stepping, conditioning, and data extraction can eliminate nominal quantum speedups [5]. The proposed experiments measure each of these costs and include a no-go decision if no crossover survives.

2 Benchmark Definition

2.1 Governing equations

For velocity $\mathbf{u} = (u, v)$, pressure p , density ρ , and kinematic viscosity ν , the two-dimensional incompressible Navier-Stokes equations are

$$\nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\rho^{-1} \nabla p + \nu \nabla^2 \mathbf{u}. \quad (2)$$

The exact convecting Taylor-Green solution is

$$u(x, y, t) = U_c + V_0 \sin \xi \cos \eta A(t), \quad (3)$$

$$v(x, y, t) = V_c - V_0 \cos \xi \sin \eta A(t), \quad (4)$$

$$p(x, y, t) = p_0 + \frac{\rho V_0^2}{4} [\cos(2\xi) + \cos(2\eta)] A(t)^2, \quad (5)$$

where

$$\xi = \frac{x - U_c t}{L_c}, \quad \eta = \frac{y - V_c t}{L_c},$$

$$A(t) = \exp\left(-\frac{2\nu t}{L_c^2}\right). \quad (6)$$

We use

$$L_{\text{box}} = 2\pi, \quad L_c = 1, \quad V_0 = U_c = \rho = 1,$$

$$V_c = p_0 = 0, \quad \nu = \frac{V_0 L_c}{Re}. \quad (7)$$

This distinction is essential: using $L = 2\pi$ simultaneously as the domain length and trigonometric scale

makes the stated field non-periodic. The final submission will record the convention in every dataset and request organizer confirmation.

2.2 Accuracy metrics

The primary error is the relative velocity-field norm

$$\epsilon_{L_2}(t) = \frac{\|\mathbf{u}_{\text{sim}}(t) - \mathbf{u}_{\text{exact}}(t)\|_2}{\|\mathbf{u}_{\text{exact}}(t)\|_2}. \quad (8)$$

Because the uniform streamwise component can dilute the vortex error, we also report the same norm after subtracting (U_c, V_c) . The kinetic energy is

$$K(t) = \frac{1}{2} \langle u^2 + v^2 \rangle. \quad (9)$$

Mass drift, momentum drift, and $\|\nabla \cdot \mathbf{u}\|_2$ complete the physical diagnostics.

3 Classical D2Q9 Foundation

The classical and quantum paths use the same D2Q9 finite-velocity model so that the initial comparison does not mix algorithmic and discretization errors. The velocities and weights are

$$\mathbf{e}_i \in \{(0, 0), (\pm 1, 0), (0, \pm 1), (\pm 1, \pm 1)\}, \quad (10)$$

$$w_0 = \frac{4}{9}, \quad w_{1:4} = \frac{1}{9}, \quad w_{5:8} = \frac{1}{36}, \quad c_s^2 = \frac{1}{3}. \quad (11)$$

Macroscopic fields are reconstructed by

$$\rho = \sum_{i=0}^8 f_i, \quad \rho \mathbf{u} = \sum_{i=0}^8 f_i \mathbf{e}_i. \quad (12)$$

The second-order equilibrium is

$$f_i^{\text{eq}} = w_i \rho \left[1 + \frac{\mathbf{e}_i \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right], \quad (13)$$

followed by BGK collision and periodic streaming,

$$f_i^*(\mathbf{x}, t) = f_i(\mathbf{x}, t) - \omega (f_i - f_i^{\text{eq}}), \quad (14)$$

$$f_i(\mathbf{x} + \mathbf{e}_i, t + \Delta t) = f_i^*(\mathbf{x}, t), \quad \omega = \tau^{-1}. \quad (15)$$

The in-repository BGK implementation is a controlled baseline, not a state-of-the-art comparator. A Fourier pseudo-spectral or high-order finite-volume solver will be added before final advantage assessment.

4 Quantum Collision and Streaming

4.1 Carleman linearization

Near reference density ρ_0 , Equation (13) is expressed as

$$\mathbf{f}^{\text{eq}} \approx L\mathbf{f} + Q : (\mathbf{f} \otimes \mathbf{f}), \quad (16)$$

where $L \in \mathbb{R}^{9 \times 9}$ and $Q \in \mathbb{R}^{9 \times 9 \times 9}$. Define

$$\mathbf{z} = \begin{bmatrix} \mathbf{f} \\ \mathbf{f} \otimes \mathbf{f} \end{bmatrix} \in \mathbb{R}^{90}, \quad R = (1 - \omega)I + \omega L. \quad (17)$$

At order two, the collision is approximated by the linear recurrence

$$\mathbf{z}^* = M\mathbf{z}, \quad M = \begin{bmatrix} R & \omega Q \\ 0 & R \otimes R \end{bmatrix}. \quad (18)$$

Here Q in the upper-right block denotes the matrixed contraction from 81 to 9 components. General convergence of Carleman truncation is conditional; dissipative quadratic systems require explicit parameter bounds and cannot be assumed efficient for arbitrary flow regimes [3].

4.2 Unitary dilation and success probability

Let $\alpha \geq \|M\|_2$ and $A = M/\alpha$. Padding the system dimension from 90 to 128 permits an eight-qubit unitary dilation

$$U_A = \begin{bmatrix} A & \sqrt{I - AA^\dagger} \\ \sqrt{I - A^\dagger A} & -A^\dagger \end{bmatrix}. \quad (19)$$

Post-selecting the signal ancilla in $|0\rangle$ applies A to the system register. The measured success probability for the current test state is

$$p_{\text{succ}} = \frac{\|M\mathbf{z}\|_2^2}{\alpha^2 \|\mathbf{z}\|_2^2} = 0.01774, \quad (20)$$

which implies about $1/p_{\text{succ}} = 56.4$ raw attempts or an amplitude amplification scale $1/\sqrt{p_{\text{succ}}} = 7.51$. These factors are part of the resource model. QSVT and oblivious amplitude amplification provide the mathematical framework for replacing dense dilation with structured block encodings [4].

4.3 Coherent streaming and extraction

For a periodic $N_x \times N_y$ grid, the position registers require $\lceil \log_2 N_x \rceil + \lceil \log_2 N_y \rceil$ qubits. Direction-controlled modular addition implements Equation (15) as a permutation. The primary outputs are $K(t)$ and selected Fourier coefficients. Full-field tomography is not included in the claimed advantage route because its output cost grows with the number of degrees of freedom.

The current spatial validator reconstructs $\mathbf{f} \otimes \mathbf{f}$ classically between timesteps. This is useful for physics validation but is not an end-to-end quantum algorithm. The central research milestone is a coherent local/global lift or an alternative incompressible-LBM construction that preserves a bounded advantage [5, 6].

5 Preliminary Results

5.1 Corrected classical datasets

The legacy datasets used the inconsistent length convention and produced approximately 25–33% relative

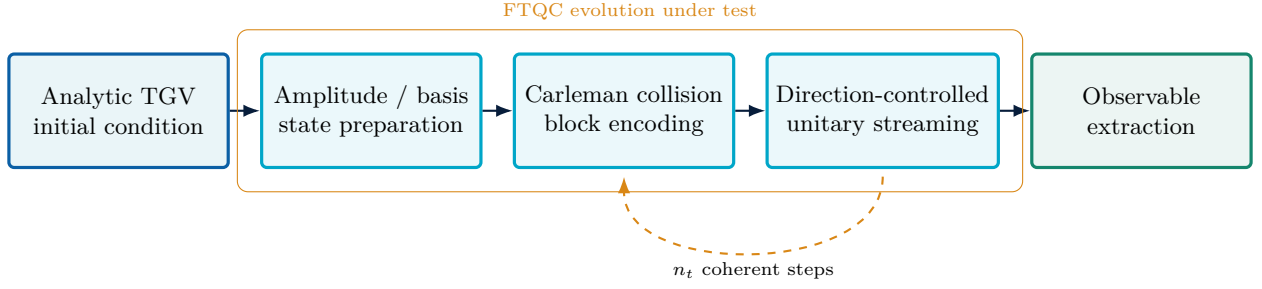


Figure 1: Proposed end-to-end method. The claimed quantum path must keep the lifted state coherent between collision and streaming; classical re-lifting is allowed only in validation experiments, not in the advantage claim.

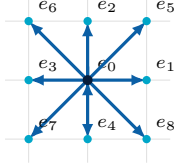


Figure 2: D2Q9 velocity stencil used by both solver paths. Streaming is a permutation and is therefore naturally unitary on a periodic grid.

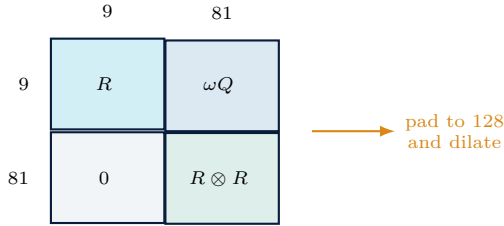


Figure 3: Structure of the 90-dimensional local order-two Carleman collision. The production objective is to compile this structure rather than synthesize a generic dense 256×256 unitary.

error. They were regenerated with $L_{\text{box}} = 2\pi$, $L_c = 1$, and $t = 1$. Final errors now lie between 4.43×10^{-3} and 4.63×10^{-3} for the configured scaling points.

5.2 Collision, circuit, and noise evidence

The local block encoding reproduces the 90-dimensional collision with maximum amplitude error 6.94×10^{-17} , and Equation (19) is unitary to 5.11×10^{-15} . A full spatial comparison at $(\text{Re}, N, t) = (100, 32, 0.1)$ gives Carleman-versus-exact-BGK relative errors 1.28×10^{-5} in velocity and 2.21×10^{-7} in populations.

The corrected streaming noise experiment transpiles to the $\{u, \text{CX}\}$ basis and actually passes a depolarizing noise model to Aer. With 20,000 shots, one-qubit error 10^{-3} , and two-qubit error 10^{-2} , the noisy-versus-ideal Hellinger fidelity is 0.964 and the total variation distance is 0.036.

These counts include routing but exclude arbitrary-rotation synthesis into Clifford+ T , error correction, and magic-state factories. They are a dense upper bound, not a favorable resource claim.

Table 1: Measured corrected classical scaling points at $t = 1$. The Re = 5000 row is an under-resolved pilot, not a production result.

Re	N	Steps	Runtime (s)	ϵ_{L_2}
10	32	395	0.86	0.004635
100	64	789	4.53	0.004432
400	128	1578	51.99	0.004497
1000	256	3156	509.83	0.004526
5000	128	1578	65.50	0.004533*

*Measured pilot; grid convergence not established.

Table 2: Connectivity-aware resource proxy for the complete local dense collision dilation on an eight-qubit bidirectional ring.

Resource	Measured value
Transpiled depth	107,128
Single-qubit u gates	51,483
CX gates	87,216
Raw post-selection probability	0.01774
Expected raw attempts	56.4
Amplitude amplification scale	7.51

6 Experimental Program

The experiments are designed around fixed physical accuracy, not fixed grid size. Each configuration records code revision, seeds, precision, dependency versions, processor or backend, transpiler settings, raw counts, and wall-clock protocol.

The required final plots are: (i) time-to-solution versus Re at fixed ϵ , (ii) memory or physical qubits versus Re, (iii) L_2 and kinetic-energy error versus Re, and (iv) a stacked cost decomposition for preparation, collision, streaming, amplification, and measurement.

7 End-to-End Advantage Test

A compact state representation does not prove an advantage. For one declared observable and error tolerance ϵ , total quantum time is modeled as

$$T_Q(\epsilon, \text{Re}) = T_{\text{prep}} + n_t [N_{\text{BE}} C_{\text{AA}}(p) + N_S] \times t_{\text{cycle}} + N_{\text{shots}} T_{\text{meas}}. \quad (21)$$

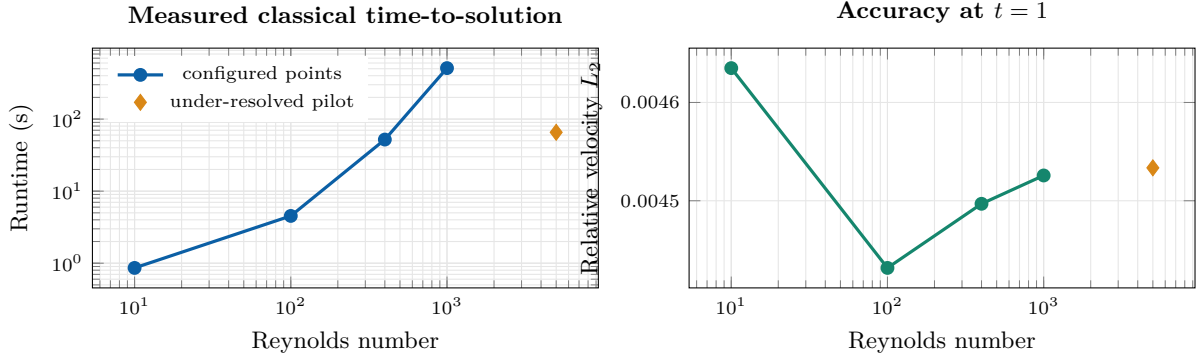


Figure 4: Directly measured corrected sweep. Runtime values are machine-specific and presently have one repetition; final submission points will use warmups, at least five repetitions, and confidence intervals.

Table 3: Planned experiments and decision criteria.

Work package	Sweep	Recorded outputs	Pass criterion
Physics convergence	Re = 10, 100, 400, 1000, 2000, 5000; $N = 32$ –2048; $Ma = 0.1$ –0.0125	Total/vortex L_2 , energy, phase, mass, momentum, divergence	Grid and Mach convergence at $t = 1$
Collision fidelity	Carleman order 1–3; 1–1000 steps; density and velocity perturbations	Population and observable drift versus exact BGK	Truncation error fits declared budget
Quantum correctness	1, 2, 5, 10, and 50 coherent timesteps	State and observable error, success probability	No classical operation in claimed path
Noise sensitivity	Gate error 10^{-4} – 10^{-2} ; shots 10^3 – 10^6	Fidelity, bias, variance, confidence interval	Noise applied after compilation
Resource scaling	$N = 8$ –2048, $Re = 10$ –5000, $\epsilon = 10^{-1}$ – 10^{-4}	Logical/physical qubits, T count/depth, block calls, shots, runtime	Complete cost ledger
Classical comparison	BGK plus spectral or high-order finite volume	Runtime, peak memory, accuracy	Same physics, observable, and tolerance

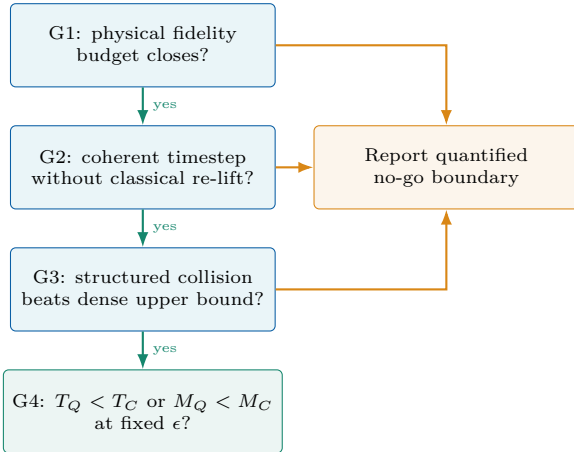


Figure 5: Evidence gates for accepting or rejecting a bounded advantage claim. Failure is reported as a quantified no-go result rather than hidden.

Here N_{BE} is block-encoding cost, $C_{AA}(p)$ is amplification overhead, N_S is streaming cost, and N_{shots} follows from the observable precision. The logical cycle time includes routing, rotation synthesis, code distance, magic-state factory throughput, and a declared failure probability.

The claim is accepted only when total time or mem-

ory is lower than both classical comparators at the same (Re, t, ϵ) for the same observable. Three hardware levels will be reported: ideal logical all-to-all, routed logical nearest-neighbor, and fault tolerant with physical qubits and wall time.

8 Risks and Mitigations

9 Schedule and Deliverables

The proposed sixteen-week program has six gated phases:

- Weeks 1–2: benchmark lock.** Complete convergence study, provenance schema, and independent classical comparator.
- Weeks 3–6: coherent algorithm.** Remove classical re-lifting from the claimed path and validate several timesteps.
- Weeks 7–10: circuit engineering.** Construct the sparse block encoding and compile collision plus streaming.
- Weeks 11–13: scale study.** Execute Re and accuracy sweeps and produce fixed-error scaling figures.

Table 4: Principal technical risks.

Risk	Likelihood / impact	Mitigation
Carleman truncation fails at high Re	High / high	Stability map, rescaling, adaptive order, and pivot to an end-to-end bounded-advantage incompressible LBM formulation
Post-selection dominates	High / high	Optimize normalization, exploit local encoding, include amplification, and enforce a stop gate
Dense synthesis remains exponential	High / high	Exploit D2Q9 sparsity, symmetry, repeated coefficients, and local collision reuse
Data extraction removes the speedup	High / high	Restrict primary outputs to kinetic energy and selected modes; account for every sample
Classical reference is weak	Medium / high	Add pseudo-spectral or high-order finite-volume baselines at identical tolerance
Benchmark convention is disputed	Medium / medium	Publish both length scales and obtain organizer confirmation

5. **Weeks 14–15: FTQC assessment.** Estimate Clifford+ T , code distance, factories, physical qubits, and wall time.
6. **Week 16: submission.** Deliver the paper, source, code, corrected datasets, raw results, and independent rerun checklist.

10 Expected Contribution

The project is designed to produce a scientifically useful result in either direction. A successful outcome is a defensible bounded advantage for selected aerodynamic observables, supported by a complete resource ledger. A negative outcome is a quantified boundary identifying whether truncation, conditioning, amplification, synthesis, or readout prevents a crossover. In both cases, the corrected Taylor-Green benchmark, nine-population collision model, and reproducibility artifacts provide a foundation for subsequent Airbus quantum-CFD evaluations.

11 Conclusion

This proposal advances beyond a streaming-only or toy-collision demonstration. It supplies a verified local D2Q9 Carleman collision block, corrected periodic datasets, full spatial fidelity checks, an applied-noise experiment, and routed resource evidence. The same evidence also shows why advantage cannot yet be claimed: dense collision synthesis and low post-selection probability are costly, and coherent multi-step lifting remains open. The proposed program turns those limitations into explicit experiments and decision gates. Only an end-to-end fixed-accuracy crossover for selected observables will be reported as quantum advantage.

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